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Chapter 10

Verification and Validation

Verification and validation are independent procedures and processes that are used together for checking the quality of software engineering products. Software configuration management (SCM) builds, services, and systems engineering must meet requirements and specifications that fulfill intended purposes. The words "verification" and "validation" are prefixed by "independent" indicating that verification and validation are to be performed by software quality engineering (SQE). "Independent verification and validation" can be abbreviated as "IV&V."

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10.1 Verification

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Verification is intended to check that a software engineering product meets a set of software design specifications. In the software development phase, verification procedures involve special tests to model or simulate software products performing a review or analysis of the modeling and buy-off results. Verification plans and procedures involve regular repeating tests devised specifically to ensure that the software engineering product, service, or system continues to meet the initial software design requirements, specifications, and regulations. It is a process that is used to evaluate whether a product, service, or system complies with regulations and conditions imposed at the start of a software engineering development phase. Verification can be conducted in stages of software development, scale-up, or production as an internal process.

Verification of software and equipment usually consists of design qualification (DQ), installation qualification (IQ), operational qualification (OQ), and performance qualification (PQ).

Verification is performed by an SQE by confirming through reviews and testing that the software and equipment meets the written acquisition specification and requirements. The relevant plans, documents, and manuals of software/equipment are provided to be thoroughly performed by the software users who work in regulatory environments.

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10.2 Validation

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Validation is intended to ensure a software product, service, or system results meet the operational needs and expectations. Validation plans and procedures involve modeling using simulations to predict faults or gaps that might lead to invalid or incomplete verification or development of a software product, service, or system.

Validation requirements, specifications, and regulations are used as a basis for qualifying a software development flow or verification flow for a software product, service, or system. Additional validation plans and procedures are also designed specifically to ensure that modifications made to an existing qualified development flow or verification flow will have the effect of producing a software product, service, or system that meets the initial design requirements, specifications, and regulations. Validation helps to keep the flow qualified and its processes establishing evidence that provides a high degree of quality assurance that a product, service, or system accomplishes its intended requirements. This often involves acceptance of fitness for purpose with software engineers and other product stakeholders. It is entirely possible that a software product passes when verified, but fails when validated. This can happen when a product is built as per the specifications, but the specifications themselves fail to address the software engineering needs.

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10.3 Verification and Validation Test Plan

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The verification and validation test plan describes how software engineering products will be tested and defines the items that are to be tested. These items may be the requirements documentation or software engineering design documentation. There could be different names for these documents, but no matter your testing comes out of these documents. You have to figure out not only what tests you need to run but also what procedures need to be in place for the test. A test has planned inputs, so think about what the expected results are based on those planned inputs.

Software testing consists of various tasks, such as defining testing schedules and creating test cases that need to be completed in order to carry out the compliant testing. Elements of the testing tasks are testing plans, test design specifications, test cases, test procedures, test results, and defects. The details in a specific environment in which the test is conducted will show how the solution works in the environment. Look into the nonfunctional requirements to determine test needs such as software test, configuration test, and so on.

Knowing where the tests are conducted is important to developing the test plan. You want to consider the following:

- If the test is in a lab, know the lab conditions.
- Check environmental concerns that could affect the test.
- Test areas must be accessible to the testers for verification and validation.

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10.4 Software Engineering Verification and Validation

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Software engineering verification and validation is the process of investigating whether a software system satisfies specifications and standards and it fulfills the required expectations. The scopes of verification and validation are as follows:

- *Verification*: Are we building the product right?
- *Validation*: Are we building the right product?

Software engineering verification is the process of checking whether a software engineering achieves its goal without any issues and concerns. This ensures products developed are right. It verifies whether the developed product fulfills the requirements and expectations.

Verification is static testing of activities involved in verification through

- Inspections.
- Reviews.
- Walkthroughs.
- Desk checking.

Software engineering validation is the process of checking whether the software engineering product is up to the performance or in other words has high-level requirements. It is the process of checking whether software engineering products are developed in the right way, and validations of actual and expected products are carried out by Dynamic Testing. Activities involved in validation are

- Black box testing.
- White box testing.
- Unit testing.
- Integration testing.

Verification and validation are shown in Figure 10.1.

Figure 10.1 Verification and validation.

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Summary

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My third book titled *Effective Processes for Quality Assurance* will describe the effective processes and activities performed in support of verification and validation. The purpose of verification is to ensure that software products meet its specified requirements, which is accomplished through various activities ranging from in-process reviews and audits. The purpose of validation is to ensure the work product fulfills its intended use when placed in its intended environment. SQE representatives will audit all activities and support qualification by monitoring all activities. The following tasks and activities must be performed:

- Analyze data from verification and validation to show expected results.
- Ensure correct standards of products submitted for verification and validation.
- Document all results of each test activity.
- Coordinate verification and validation results with software team members.

After verification and validation, reports are developed and written, and requirements are determined, prepare compliance for closure to plans, procedures, and documentation to show proof of compliance and to be ready for release.

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Further Reading

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